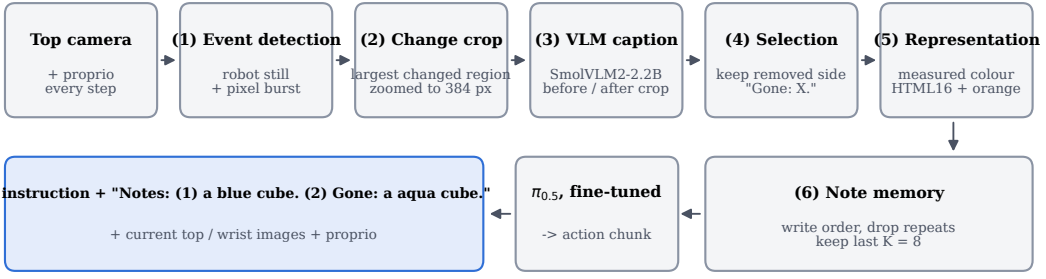
Top camera
+ proprio
every step
(1) Event detection
robot still
+ pixel burst
(2) Change crop
largest changed region
zoomed to 384 px
(3) VLM caption
SmolVLM2-2.2B
before / after crop
(4) Selection
keep removed side
"Gone: X."
(5) Representation
measured colour
HTML16 + orange
(6) Note memory
write order, drop repeats
keep last K = 8
$\pi_{0.5}$, fine-tuned
-> action chunk
instruction + "Notes: (1) a blue cube. (2) Gone: a aqua cube."
+ current top / wrist images + proprio